Cross-Modality Domain Adaptation Based on Semantic Graph Learning: From Optical to SAR Images

Xiufei Zhang, Zhongling Huang, *Member, IEEE*, Xiwen Yao, *Member, IEEE*, Xiaoxu Feng, Gong Cheng, *Senior Member, IEEE*, and Junwei Han, *Fellow, IEEE*

***Abstract*— Synthetic aperture radar (SAR) imaging provides a distinct advantage in scene understanding due to its capability for all-weather data acquisition. However, in comparison to easily annotated optical remote sensing images (RSIs), the lower imaging quality of SAR images presents significant challenges in obtaining manually annotated training data, which poses substantial issues for SAR image analysis. In this article, we employ the domain adaptation (DA) that leverages labeled optical images to better understand unlabeled SAR images. Global feature alignment as a method for DA has demonstrated effectiveness in transferring knowledge, yet it faces challenges in cross-modality adaptation from optical remote sensing to SAR images due to their differing imaging mechanisms. With distinct visual features between optical and SAR images, the semantic dependence is difficult to construct, which results in low-quality pseudolabel assignment for SAR images. To address the above issue, we propose a semantic graph learning framework (SGLF) to comprehensively align the global features of optical remote sensing and SAR images by modeling the cross-modality semantics and generating high-quality pseudolabels. It can be applied for SAR scene classification and object detection when only optical RSIs are labeled. Specifically, a cross-modality semantic graph alignment (CSGA) module is constructed to model and align the second-order semantic dependencies by aggregating cross-modality visual semantic information. Then, an uncertainty-based robust pseudolabel generation (URPG) module is designed to generate pseudolabels for effective semantic alignment and self-training by modeling the uncertainty of pseudolabels for each SAR image. Comprehensive experiments show that our proposed method outperforms the state-of-the-art methods on scene classification (NWPU-RESISC45 → WHU-SAR6, MLRSNet → NWPU-SAR6, MLRSNet → NWPU-SAR6, and NWPU-RESISC45 → NWPU-SAR6) and object detection (MASATI-ship → SSDD, MVSRD → SARDet-vehicle, and DIOR-airplane → SAR-airplane) tasks. The code and datasets are publicly accessible at https://github.com/XZhang878/SGLF**

***Index Terms*— Cross-modality semantic graph alignment (CSGA), domain adaptation (DA), semantic graph learning, synthetic aperture radar (SAR) images, uncertainty-based robust pseudolabel generation (URPG).**

Received 14 December 2024; revised 7 March 2025; accepted 8 April 2025. Date of publication 11 April 2025; date of current version 28 April 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62071388, Grant 62136007, Grant U20B2065, and Grant 62036005; and in part by the Key Research and Development Program of Shaanxi Province under Grant 2023-YBGY-224. *(Corresponding authors: Zhongling Huang; Xiwen Yao.)*

Xiufei Zhang, Zhongling Huang, Xiwen Yao, Gong Cheng, and Junwei Han are with the School of Automation, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: huangzhongling@nwpu.edu.cn; yaoxiwen517@gmail.com).

Xiaoxu Feng is with the Hefei Comprehensive National Science Center, Hefei 230026, China.

Digital Object Identifier 10.1109/TGRS.2025.3559915

I. Introduction

With the swift progress in sensor and imaging technologies, synthetic aperture radar (SAR) has gained extensive use across military reconnaissance and civilian sectors, leveraging its all-weather and continuous capabilities for Earth observation and imaging [1]. Benefiting from the surge in the abundance and excellence of SAR images and the robust expansion of convolutional neural network (CNN) applications, a wide array of CNN-based approaches for classification and detection have been developed, delivering highly promising results [2], [3], [4].

Although these CNN-based methods have shown remarkable results on SAR images, SAR images require a large amount of labeled information. In particular, the annotation of SAR images is more challenging compared to easily annotated optical remote sensing images (RSIs) due to the different imaging mechanisms. SAR images usually exhibit speckle patterns as a result of radar scattering from various surfaces and polarizations, which typically leads to lower imaging quality. Nonetheless, SAR sensors provide all-day, all-weather observation capabilities, offering critical advantages over optical remote sensing technologies, particularly in conditions such as nighttime or cloudy weather where optical systems are ineffective [5]. In some practical scenarios, SAR images are the only source of data available, necessitating their use for object and terrain interpretation. This work focuses on addressing such scenarios. One of the main challenges with SAR imagery is the difficulty in large-scale manual annotation due to its unique electromagnetic scattering characteristics, which significantly differ from optical imagery. Therefore, it is necessary to study the analysis of SAR images. Moreover, there are essential differences between the characteristics of SAR and optical RSIs. For instance, SAR features are susceptible to viewing angle, parameters, and so on, whereas optical features are not influenced by these variables [6], [7], [8]. The discrepancy in data distribution between optical RSIs and SAR images, which is perceived as a domain shift, is an inevitable occurrence. This discrepancy disastrously limits the enhancement of model generalization. Therefore, when a model trained on a labeled optical remote sensing dataset (source domain) by using a CNN-based approach is extended to an unlabeled SAR dataset (target domain), the performance of the model may fail. Based on that, it is essential to develop an accurate method for remote sensing imagery that effectively addresses the domain gap.

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.

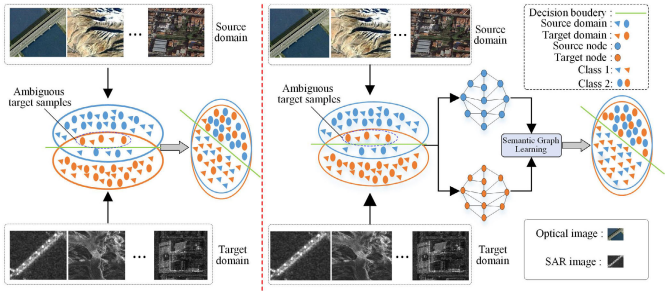


Fig. 1. Motivation of the proposed method. (Left) Global feature alignment is a challenge due to the large visual differences between optical and SAR images, and the semantic dependencies are difficult to construct, which leads to low-quality pseudolabel assignments for ambiguous SAR images. (Right) Our method constructs semantic dependencies via semantic graph learning and introduces high-quality pseudolabels as category information to guide knowledge transfer from optical to SAR domains.

To improve the generalization performance of deep learning models on the target domain, an attractive and feasible approach to tackle this problem is by using the domain adaptation (DA) approach. Recently, DA has achieved great success in natural images, which aims to transfer knowledge from the labeled information domain to the unlabeled information domain by reducing distribution differences [9], [10]. Following these methods, DA has been widely utilized between optical RSIs and SAR images. In recent years, there has been a surge of research on DA between optical and SAR images. Shi et al. [11] proposed a DA framework based on progressive transfer for SAR ship detection by transferring knowledge from the optical domain to the SAR domain. Zhang et al. [12] constructed a hierarchical similarity alignment strategy for cross-domain SAR object detection to explore knowledge transfer from the optical domain to the SAR domain. Yuan et al. [13] designed an OmniAdapt framework to gradually reduce the distance between the optical and SAR domains from the perspectives of pixel, feature, and classifier levels, achieving knowledge transfer from optical to SAR domains. Shi et al. [14] presented a DA method that effectively transfers knowledge from labeled optical RSIs to unlabeled SAR images by progressively aligning domain and class-level features, significantly enhancing ship classification performance. Although the above methods have achieved great success, these studies specifically concentrate on aligning global feature distributions. As shown in Fig. 1, global feature alignment can result in low-quality pseudolabel [15], [16] assignment for the ambiguous SAR images in the process of DA due to the semantic dependence between the visual features of optical images and SAR images being difficult to construct. The proposed method leverages semantic graph learning to establish semantic dependencies and incorporates high-quality pseudolabels to facilitate knowledge transfer from optical to SAR domains.

In addition, our method does not require the registration of optical RSIs and SAR images. Image registration seeks to establish correspondences between images, even when they exhibit visual or structural discrepancies. This often involves applying geometric transformations, affine transformations, or more complex mapping functions [17], [18]. In this article, the proposed method emphasizes feature-level alignment rather than image-level alignment. Unlike approaches that require precise pixel registration between optical and SAR images [19], our method focuses on understanding shared semantic concepts, such as identifying objects such as ships, across different SAR and optical domains. This focus on semantic consistency allows the method to remain robust against distortions and variations present in the images, as these have minimal impact on the feature-level understanding within our framework. In this case, there is no significant impact on the performance of the DA model for distortion and scattering of SAR images.

To overcome the above challenge, we propose a semantic graph learning framework (SGLF) to comprehensively align the deep semantic features of optical remote sensing and SAR images by modeling the cross-modality semantics and generating high-quality pseudolabels. The SGLF consists of two modules, namely, cross-modality semantic graph alignment (CSGA) and uncertainty-based robust pseudolabel generation (URPG) modules. The CSGA module models and aligns second-order semantic dependencies by aggregating cross-modality visual semantic information. Specifically, we first generate the semantic interaction graph by explicitly interacting the semantic graph between optical and SAR images with the cross-attention mechanism. Next, the cross-modality semantic discrimination ability is effectively enhanced by aggregating both similarity information and semantic graph features. Based on this, we obtain k predictions. Finally, the cross-modality semantic is matched by constructing the differentiable top-k-based graph alignment from optical to SAR images. Accordingly, the CSGA module can achieve semantic alignment from optical images to SAR images to train a robust model.

To further strengthen the alignment of semantic features of optical and SAR images, we introduce category information into semantic feature alignment. The URPG module selects the high-quality pseudolabels as surrogates by modeling the uncertainty of the correct labeling for each SAR sample. Specifically, we generate the coarse pseudolabels for each SAR image through the designed CNN network. Meanwhile, the uncertainty scores are obtained by modeling a beta mixture model (BMM) to fit the coarse pseudolabels. Based on that, the category information is introduced into the CSGA module, and the semantic features of the same category in the optical image and SAR image are aligned. Furthermore, a robust pseudolabel learning loss (RPLL) is designed to realize domain knowledge transfer from the optical to SAR domain by selecting high-quality pseudolabels, which alleviates the representation bias toward optical-annotated images.

The collaboration between CSGA and URPG establishes a novel SGLF that can be learned end-to-end and achieves effective adaptation to the DA scenario from optical to SAR imagery. To sum up, the main contributions of the proposed framework are summarized in the following.

1) We propose a novel cross-modality DA framework from optical to SAR domains, which improves the global

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.

feature alignment by enhancing and aligning the deep semantic features of the different modalities.

2) We design the CSGA and URPG modules. The CSGA module can effectively model and align second-order semantic dependencies from optical to SAR domains by aggregating semantic graph information. The URPG module can precisely select high-quality pseudolabels by modeling the uncertainty of the correct labeling for each SAR sample.
3) We construct seven cross-modality DA datasets from optical to SAR images including scene classification and object detection tasks. The extensive experiments demonstrate that the proposed method outperforms the state-of-the-art approaches and illustrate its general applicability on different cross-modality DA tasks.

II. Related Works

A. DA for Image Classification

Following the underlying assumption [20], a wide range of DA methods [21], [22] have been proposed, aiming at learning transferable features to reduce cross-domain discrepancy. DA is often considered a promising remedy to the difficulties caused by the lack of domain-specific training data. For DA, a typical solution is to map the embedded feature distributions of the source and target domains separately to the common space for alignment, such as maximum mean discrepancy [21] and central moment discrepancy [23]. Influenced by the successful application of generative adversarial net [24], a great quantity of DA works [22], [25] have utilized adversarial learning to develop domain-invariant representations through the interplay between generator and discriminator.

Nowadays, the above DA methods have achieved great success in the field of natural images; however, these approaches are only applied to natural scenes without considering the unique challenges to RSIs [26], [27]. Compared to natural images, RSIs usually show large interdomain and intradomain variances [28]. In practice, DA in RSIs has received wider attention and research to reduce domain shift. Dominant methods such as the method in [29] designed an asymmetric adaptation neural network for DA classification in RSIs, Deng et al. [30] proposed a scale adaptive framework for DA in land cover classification, Wang et al. [31] introduced a DA semantic segmentation dataset to promote large-scale land-cover mapping, Lang et al. [32] proposed a dynamic joint correlation alignment network to conduct semisupervised heterogeneous DA for ship classification in SAR imagery, and He et al. [33] proposed a task-driven DA transfer learning method for SAR target recognition. Xu et al. [34] presented a discriminative adaptation regularization framework-based transfer learning method for ship classification in SAR images. However, these methods only focus on global feature alignment and ignore the matching of deep semantic modality information.

B. DA for Object Detection

In contrast, the single-domain object detection task is impressive; however, DA research for object detection tasks is relatively limited. That being said, several ideas have been explored in existing attempts. Saito et al. [35] presented an approach for object detection based on strong-weak domain alignment. Wu et al. [36] proposed a disentangled method based on vector decomposition for DA object detection. Zhao and Wang [37] proposed a task-specific inconsistency alignment on both subtasks to enhance the performance of the detector. Li et al. [38] designed a category dictionary-guided mode to improve the generalization ability of the detector.

The above method provides feasible solutions for transferring knowledge from the source to the target domain in the field of natural images. More recently, DA object detection has been valued in the field of RSIs. Following the practice of conventional wisdom, Zhu et al. [39] proposed a dual-head rectification alignment network for DA object detection in RSIs. Shi et al. [2] proposed an SAR DA object detection framework based on faster R-CNN, which strengthens the SAR object detector to alleviate domain discrepancy. Zhao et al. [40] proposed a DA transformer method to address the multisource SAR image ship object detection problem. However, these approaches fail to consider the discriminative nature between foreground and background, and the local object instances learned through the detector may be unreliable due to the scarcity of effective supervision information in the SAR domain.

III. Method

A. Overview of DA

We outline the architecture used for training, as shown in Fig. 2. In the DA setting, we take optical modality and SAR modality images as labeled source domain $\mathcal{D}_s : \{x_s^i, y_s^i\}_{i=1}^{\mathcal{N}_s}$ and unlabeled target domain $\mathcal{D}_t : \{x_t^i\}_{i=1}^{\mathcal{N}_t}$, respectively. $\mathcal{D}_s$ and $\mathcal{D}_t$ derived from the source distribution $\mathcal{P}_s$ and target distribution $\mathcal{P}_t$ ($\mathcal{P}_s \neq \mathcal{P}_t$), respectively. The optical and SAR domains share a feature extractor $\mathcal{F}$, and the feature extractor $\mathcal{F}$ extracts the corresponding input, given by $f = \mathcal{F}(x)$, and then outputs a probability distribution by a fully connected layer $\mathcal{FC}$. Therefore, the cross-entropy loss on the optical domain is calculated as follows:

$$\mathcal{L}_{\mathrm{CL}} = -\mathbb{E}_{(x_s^i, y_s^i)\sim\mathcal{D}_s}\left[y_s^i \log\left[\mathcal{FC}\left(f_s^i\right)\right]\right] \tag{1}$$

where y_s^i is the ground-truth label corresponding to the optical data x_s^i. Then, we train a feature extractor $\mathcal{F}$ to learn invariant features of the optical and SAR domains by confusing the domain discriminator D. The adversarial loss on the source and target domains is calculated as follows:

$$\min_{\mathcal{F}} \max_{D} \mathcal{L}_{\mathrm{ADL}} = \mathbb{E}_{x_s^i\sim\mathcal{D}_s} \log\left[D\left(\mathcal{F}\left(x_s^i\right)\right)\right] + \mathbb{E}_{x_t^i\sim\mathcal{D}_t} \log\left[1 - D\left(\mathcal{F}\left(x_t^i\right)\right)\right]. \tag{2}$$

Domain invariant features are generated by min–max training $\mathcal{L}_{\mathrm{ADL}}$. However, directly employing an adversarial learning mechanism could lead to instability issues in the training process. Therefore, we propose a novel SGLF framework to model and align the deep semantic dependencies with the help of robust pseudolabels by modeling visual semantic graphs, thereby alleviating the instability of adversarial learning throughout the training process.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.

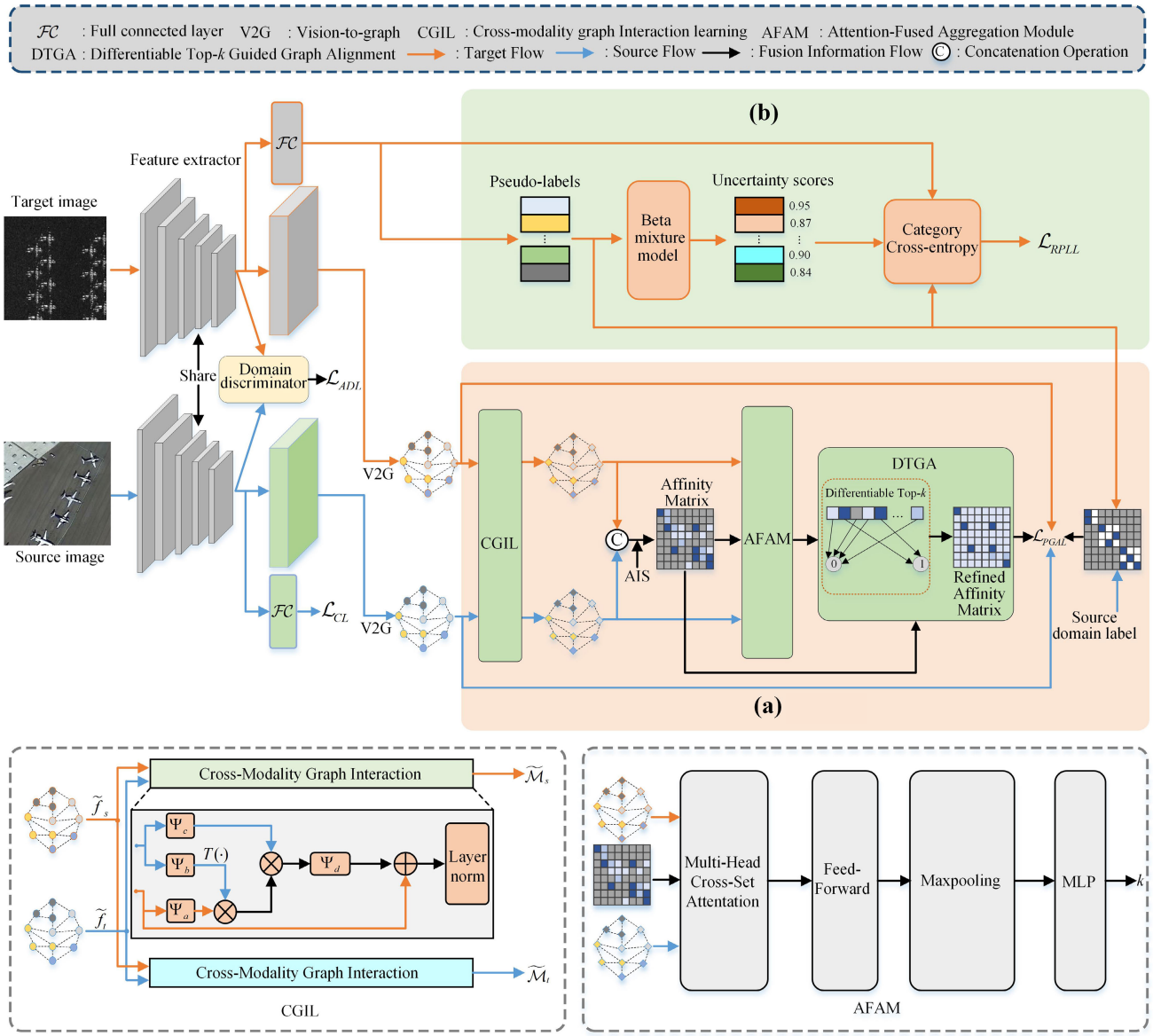


Fig. 2. Overall framework of our adaptation network. The URPG module is deployed to generate robust pseudolabels for each target-domain sample. With the help of robust pseudolabels, the CSGA module is developed to model and align the deep second-order semantic dependence features. Different colored circles and rhombuses represent different nodes. (a) CSGA. (b) URPG.

B. Cross-Modality Semantic Graph Alignment

Due to the significant differences in imaging mechanism and feature representation between optical and SAR images, therefore, we design a CSGA module to model and align deep semantic feature distribution via both semantic graph fusion and the differentiable top-k for better model generalization, as illustrated in Fig. 2.

1) Cross-Modality Graph Interaction Learning: To model the distribution of feature structures between optical modality and SAR modality images, we design a cross-modality graph interaction learning (CGIL), as shown in Fig. 2. Specifically, we utilize a graph convolutional network (GCN) [41] to transform visual features into graph space. The graph is denoted as $\Theta : \Theta = \langle \mathcal{V}, \varepsilon \rangle$, where $\mathcal{V}$ is the node set and ε is the edge set. The node $v_i, v_j \in \mathcal{V}$ correspond to the visual feature, and $e_{i,j} \in \varepsilon$ encodes the relationship between v_i and v_j nodes. The edges are denoted as $\varepsilon = [e_{i,j}]$, where $e_{i,j}$ indicates the node of the corresponding instance and expressed as follows:

$$e_{i,j} = \frac{\exp(\phi_{i,j})}{\sum \exp(\phi_{i,j})}, \quad \text{where } \phi_{i,j} = g(v_i) \cdot h(v_j)^{\mathrm{T}} \tag{3}$$

where $g(\cdot)$ and $h(\cdot)$ are two learnable fully connected layers that help to model the relation between nodes and T represents the matrix transpose. In this way, we use vision-to-graph conversion (V2G) to convert feature f into graph Θ. The node graph reasoning features $\widetilde{f}$ are described as follows:

$$\widetilde{f} = \mathrm{ReLU}(\varepsilon f \omega) \tag{4}$$

where ω is a learnable weight matrix and $\mathrm{ReLU}(\cdot)$ indicates rectified linear activation function.

Based on the above method, we establish cross-modality graph features $\widetilde{f}_s$ and $\widetilde{f}_t$ to model the local feature structure with enhanced nodes v_s and v_t. To adapt the semantic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.

structure-aware features of optical modality images and SAR modality images to better model the local feature structure distributions, we use graph features $\widetilde{f}_s$ and $\widetilde{f}_t$ for CGIL via the cross-attention mechanism

$$\widetilde{\mathcal{M}}_s = \text{softmax}\left[\left(\widetilde{f}_s\Psi_a\right)\left(\widetilde{f}_t\Psi_b\right)^{\mathrm{T}}\right]\left(\widetilde{f}_t\Psi_c\right)\Psi_d + \widetilde{f}_s \tag{5}$$

$$\widetilde{\mathcal{M}}_t = \text{softmax}\left[\left(\widetilde{f}_t\Psi_a\right)\left(\widetilde{f}_s\Psi_b\right)^{\mathrm{T}}\right]\left(\widetilde{f}_s\Psi_c\right)\Psi_d + \widetilde{f}_t \tag{6}$$

where Ψ_a, Ψ_b, Ψ_c, and Ψ_d are learnable parameters.

2) Cross-Modality Node Affinity: According to cross-modality graph interaction features $\widetilde{\mathcal{M}}_s$ and $\widetilde{\mathcal{M}}_t$, we embed the interaction features $\widetilde{\mathcal{M}}_s$ and $\widetilde{\mathcal{M}}_t$ into the AIS module to learn a relational affinity matrix to model the node correspondence between $\widetilde{\mathcal{M}}_s$ and $\widetilde{\mathcal{M}}_t$, where the AIS module consists of an affinity layer, instance normalization, and Sinkhorn layer to predict soft correspondences from the relational interaction of two modalities. In the AIS module, we utilize the affinity layer to calculate the affinity matrix $\mathcal{A}$ of the interaction graph features between two modalities as follows:

$$\mathcal{A} = \mathcal{MLP}\left[\mathcal{P}\left(\widetilde{\mathcal{M}}_s\right)\copyright\mathcal{P}\left(\widetilde{\mathcal{M}}_t\right)\right] \tag{7}$$

where $\mathcal{P}$ indicates a linear projection, $\mathcal{MLP}$ is a multilayer perceptron layer with a single output channel, and © is the concatenation operation.

To ensure that the graph affinity $\mathcal{A}$ generated by the affinity layer is all positive, we need to convert the matrix $\mathcal{A}$ into a matrix with nonnegative elements within finite values. There are currently two ways to achieve this; the most commonly used method is to use softmax to transform the rows or columns of the matrix $\mathcal{A}$. However, this method does not treat the affinity matrix as a whole when dealing with each row or column of the affinity matrix, which may lead to the problem that a smaller value in $\mathcal{A}$ turns into a larger value after conversion. We deprecate softmax and use instance normalization to transform $\mathcal{A}$. Instance normalization takes all elements into account from a global perspective and uses an exponential function to make all elements positive after transformation. To correct these outliers since there may be outliers in the affinity matrix after the instance normalization transformation, we add an extra row and an extra column to the transformed matrix and then send it to the Sinkhorn layer to calculate the soft affinity matrix $\widetilde{\mathcal{A}}$ by alternatively taking row and column normalization until the iteration converges.

3) k Prediction: Due to the rich similarity information contained in $\widetilde{\mathcal{A}}$, estimating k can be seen as a continuation of graph similarity learning. Therefore, we design an attention-fused aggregation module (AFAM) to aggregate cross-modality similarity information into node features for k prediction. Specifically, the structure of AFAM is mainly based on the transformer [34]. Taking inspiration from the multihead attention module in the transformer, we design the multihead cross-set attention to aggregate node features and affinity information from the source and target domains. We take the node features $\widetilde{\mathcal{M}}_s$ as the input query and the node features $\widetilde{\mathcal{M}}_t$ as both the input key and the value in the input key–value pair. In each head, the attention weights $\widetilde{W}_{\text{att}}$ are first computed by the input query and the key

$$\widetilde{W}_{\text{att}} = \left(\widetilde{\mathcal{M}}_s\mathcal{W}_q\right)\left(\widetilde{\mathcal{M}}_t\mathcal{W}_k\right)^{\mathrm{T}} \tag{8}$$

where $\mathcal{W}_q$ and $\mathcal{W}_k$ are learnable parameters. To aggregate similarity information $\widetilde{\mathcal{A}}$ into node features, we concatenate the affinity similarity matrix $\widetilde{\mathcal{A}}$ and $\widetilde{W}_{\text{att}}$ at a new dimension and then the concatenate matrix input into a two-layer MLP to output the final attention weights $\widetilde{W}_{\text{final}}$

$$\widetilde{W}_{\text{final}} = \text{softmax}\left(\mathcal{MLP}\begin{bmatrix}\widetilde{W}_{\text{att}}\\ \widetilde{\mathcal{A}}\end{bmatrix}\right). \tag{9}$$

The attention weight $\widetilde{W}_{\text{final}}$ is multiplied by the embedded value to output the final attention in each head

$$\widetilde{\text{Att}}_{\text{qkv}} = \widetilde{W}_{\text{final}}\left(\widetilde{\mathcal{M}}_t\mathcal{W}_v\right) \tag{10}$$

where $\mathcal{W}_v$ is the learnable parameter. We then fuse the features from all heads through concatenation and linear transformation operations. The output attention features $\widetilde{\text{Att}}_{\text{qkv}}$ are passed through a feedforward module from the transformer [34] and a max-pooled (Maxpooling) to obtain the most significant aggregated embedding features. Finally, a two-layer MLP with sigmoid activation is utilized to generate the final k-prediction so that k is obtained by rounding.

4) Top-k-Based Cross-Modality Graph Alignment: Since the graph inherently characterizes a variety of second-order dependencies, it is crucial for its ability to both model and align the local feature distributions across different modalities, leveraging the second-order knowledge resident in the edges. As a result, we extend the approach of second-order graph alignment to the entire second-order graph space, ensuring a more robust adaptation of the local feature structure. Although Li et al. [42] proposed a semantic complete graph alignment method, this method obtains a coarse affinity matrix that may contain misaligned nodes. Therefore, we construct a top-k-based graph alignment strategy to suppress the false alignment through an estimated number of inliers k. Specifically, we flatten the soft affinity matrix $\widetilde{\mathcal{A}}$ into its vectorized version $a = [a_1, a_2, a_3, \ldots, a_n]$, where n is the number of all nodes in $\widetilde{\mathcal{A}}$. To select the top-k matches from a in a differentiable manner, we, respectively, establish the elements of $a_{\max}$ with space size k and $a_{\min}$ with $n-k$. In this case, elements in $\widetilde{\mathcal{A}}$ that are aligned with $a_{\max}$ are selected, while other elements that are aligned with $a_{\min}$ are discarded. We denote u and v as the marginal distributions

$$u = \mathbf{1}_n^{\mathrm{T}}, \quad v = [n-k, k]^{\mathrm{T}} \tag{11}$$

where $\mathbf{1}$ is a vector of ones and the distance matrix E is denoted as follows:

$$E = \begin{bmatrix} a_1 - a_{\min} & a_2 - a_{\min} & \cdots & a_n - a_{\min} \\ a_{\max} - a_1 & a_{\max} - a_2 & \cdots & a_{\max} - a_n \end{bmatrix}. \tag{12}$$

Since the top-k matches can as an optimal transport (OT) problem, the OT problem is expressed as follows:

$$\min_{\overline{\Gamma}} \text{tr}\left(\overline{\Gamma}^{\mathrm{T}}E\right) + \epsilon h\left(\overline{\Gamma}\right), \quad \text{s.t. } \overline{\Gamma}\mathbf{1} = u,\ \overline{\Gamma}^{\mathrm{T}}\mathbf{1} = v \tag{13}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.

where $\overline{\Gamma} \in [0,1]^{n\times 2}$ is the transportation matrix, $\mathrm{tr}(\cdot)$ represents the trace of the matrix, ϵ is the regularization factor, and $h(\overline{\Gamma}) = \sum_{i,j}\overline{\Gamma}_{i,j}\log\overline{\Gamma}_{i,j}$. Then, we use the differentiable top-k based on the Sinkhorn algorithm [43] to optimize (13) until convergence to obtain the optimal $\overline{\Gamma}^*$. We reshape $\overline{\Gamma}^*$ to obtain the refined affinity matrix $\widetilde{\mathcal{A}}^*$.

Although we obtain a refined affinity matrix $\widetilde{\mathcal{A}}^*$, we do not know which nodes of the same category are most suitable for DA. We only know that these nodes may be in the same category. To address this issue, with the help of robust pseudolabels, we propose robust pseudolabel-guided graph alignment loss (PGAL) to ensure the intrinsic correspondence of the local feature structure between optical modality and SAR modality images. This results in an effective distribution alignment among nodes with similar semantic attributes. The PGAL is defined as follows:

$$\begin{aligned}\mathcal{L}_{\mathrm{PGAL}} = &\sum\left(\widetilde{\mathcal{A}}^* \odot \mathrm{X}^{\mathrm{gt}} - 1\right)^2 \log\widetilde{\mathcal{A}}^* \\ &+ \sum\left[\widetilde{\mathcal{A}}^* \odot \left(1-\mathrm{X}^{\mathrm{gt}}\right)\right]^2 \log\left(1-\widetilde{\mathcal{A}}^*\right) \\ &+ \sum\left(\widetilde{\mathcal{A}}^* \odot \widetilde{f}_s - \widetilde{\mathcal{A}}^* \odot \widetilde{f}_t\right) \end{aligned} \tag{14}$$

where $\odot$ indicates the elementwise multiplication, $\mathrm{X}^{\mathrm{gt}} \in \mathbb{R}^{\mathcal{B}_s\times\mathcal{B}_t}$ indicates one-hot encode matrix of both source-domain label and target-domain robust pseudolabel, and $\mathcal{B}_s$ and $\mathcal{B}_t$ are the source and target batch sizes, respectively.

C. Uncertainty-Based Robust Pseudolabel Generation

1) Uncertainty Estimate of Correct Labeling: In the SAR domain, $\widetilde{p}_t, \widetilde{y}_t = \arg\max_c \mathrm{softmax}(\mathcal{FC}(f_t^i))$, and the uncertainty probability $\widetilde{p}_t$ quantifies which samples are most confident. Since the data of the SAR domain have no label information, the uncertainty probability $\widetilde{p}_t$ and pseudolabels $\widetilde{y}_t$ may be inaccurate. We construct an uncertainty estimate of correct labeling to model the uncertainty confident probability $\widetilde{p}_t$ for pseudolabel by using BMM [49]

$$p(z) = \sum_{h=1}^{H}\pi_h p(z|h) \tag{15}$$

where z is the probability $\widetilde{p}_t$, π_h is the mixing coefficients, and $H \in \mathbb{N}$ is the number of components. The more flexible beta distribution allows modeling both symmetric and skewed distributions over [0, 1] [50]. Therefore, the BMM better approximates the uncertainty probability distribution for mixtures of correctly and wrongly labeled samples. The beta distribution over z is defined as follows:

$$p(z|\alpha,\beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} z^{\alpha-1}(1-z)^{\beta-1} \tag{16}$$

where $\alpha, \beta > 0$, $\Gamma(\cdot)$ is the Gamma function, and the importance of confidence per sample is given by substituting the above (16) into (15). Here, we set $H = 2$ indicating the correctly and wrongly labeled components. We use the expectation–maximization (EM) algorithm [50] to fit the BMM, by which we obtain the posterior probability of correct labeling

$$\gamma_h(z) = p(h|z) = \frac{\pi_h p(z|\alpha_h,\beta_h)}{\sum_{i=1}^{H}\pi_i p(z|\alpha_i,\beta_i)}. \tag{17}$$

In the M-step, fixing the posterior $\gamma_h(z)$, we estimate the distribution parameters α_h and β_h by using method of moments

$$\alpha_h = \bar{z}_h\left(\frac{\bar{z}_h(1-\bar{z}_h)}{s_h^2} - 1\right), \quad \beta_h = \frac{\alpha_h(1-\bar{z}_h)}{\bar{z}_h} \tag{18}$$

where $\bar{z}_h$ is the weighted average of the probability $\widetilde{p}_t$ corresponding to each training target-domain sample and s_h^2 is a weighted variance estimate

$$\bar{z}_h = \frac{\sum_{i=1}^{\mathcal{N}_t}\gamma_h(z_i)z_i}{\sum_{i=1}^{\mathcal{N}_t}\gamma_h(z_i)} \tag{19}$$

$$s_h^2 = \frac{\sum_{i=1}^{\mathcal{N}_t}\gamma_h(z_i)\left(z_i-\bar{z}_h\right)^2}{\sum_{i=1}^{\mathcal{N}_t}\gamma_h(z_i)} \tag{20}$$

$$\pi_h = \frac{1}{\mathcal{N}_t}\sum_{i=1}^{\mathcal{N}_t}\gamma_h(z_i). \tag{21}$$

Finally, we obtain the uncertainty estimate of a sample being correct or wrong labeling through the posterior probability

$$p(h|z) = \frac{p(h)p(z|h)}{p(z)} \tag{22}$$

where $p(h=1|z)$ denotes the importance that the sample is correctly labeled.

2) Robust Pseudolabel Learning: In this section, we design an RPLL to mitigate the representation bias toward the optical-annotated image, which enhances the robustness of pseudolabeling for SAR images. Here, we set a random variable $h_i \in \{0,1\}$ with values of 1 and 0 to indicate whether the SAR image is correctly labeled or incorrectly labeled. We denote the probability of correct labeling as $P_t(h_i = 1|x_t^i, \widetilde{y}_t)$, and then, our RPLL is defined as follows:

$$\mathcal{L}_{\mathrm{RPLL}} = \frac{1}{\sum_{i=1}^{\mathcal{N}_t}\mathcal{W}(x_t^i)}\sum_{i=1}^{\mathcal{N}_t}\mathcal{W}\left(x_t^i\right)\mathcal{H}\left(\left(\mathcal{FC}\left(f_t^i\right)\right),\widetilde{y}_t\right) \tag{23}$$

where $\mathcal{H}(\cdot,\cdot)$ is the category cross-entropy (CCE) [51]. $\mathcal{W}(x_t^i)$ is defined based on confidence score of correct labeling

$$\mathcal{W}\left(x_t^i\right) = \begin{cases}\eta_i, & \text{if } \eta_i > 0.5 \\ 0, & \text{otherwise}\end{cases} \tag{24}$$

where $\eta_i = P_t(h_i = 1|x_t^i, \widetilde{y}_t)$. In this way, low-quality target data with a probability of correct labeling less than 0.5 are discarded to improve correct labeling quality.

D. Model Optimization

The cooperation of CSGA and URPG establishes an end-to-end SGLF framework to dynamically align deep semantic features and enhance the discriminability of pseudolabels. Therefore, the overall optimization objective of the proposed framework is denoted as

$$\mathcal{L} = \mathcal{L}_{\mathrm{CL}} + \mathcal{L}_{\mathrm{ADL}} + \lambda_1\mathcal{L}_{\mathrm{PGAL}} + \lambda_2\mathcal{L}_{\mathrm{RPLL}} \tag{25}$$

where λ_1 and λ_2 are hyperparameters that balance the loss terms.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.

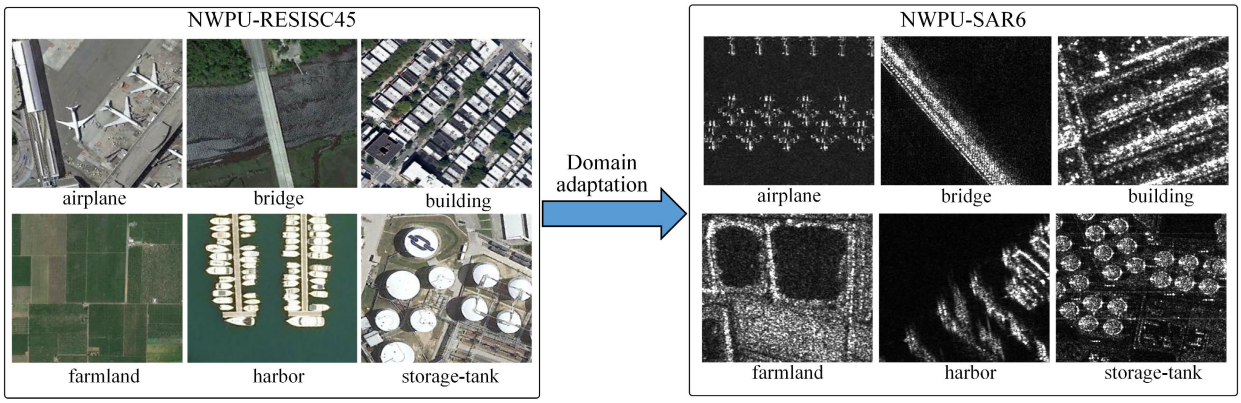


Fig. 3. Some examples are taken from the NWPU-RESISC45 and NWPU-SAR6 datasets, respectively. The NWPU-RESISC45 and NWPU-SAR6 datasets consist of six categories: airplane, bridge, building, farmland, harbor, and storage tank. We construct a cross-modality DA for knowledge transfer from the NWPU-RESISC45 (source domain) to the NWPU-SAR6 (target domain) datasets.

TABLE I
ACCURACY (%) ON NWPU-RESISC45 → WHU-SAR6 FOR DA

Methods	bridge	mountain	building	lake	farmland	river	Avg.
ResNet-50 [44]	19.60	89.5	64.00	56.75	59.29	29.25	53.06
DANN+E [45]	76.00	95.50	74.50	54.25	28.00	68.25	66.08
CDAN+E [22]	69.60	95.00	69.00	37.75	**88.00**	**73.50**	72.14
AFN [9]	16.00	99.00	85.75	45.25	77.65	53.50	62.86
CDAN+BNM [46]	28.40	98.75	72.25	35.25	71.53	59.50	60.95
ATM [10]	60.40	98.75	84.00	88.47	45.75	58.25	72.60
CGDM [47]	87.60	99.75	96.50	54.50	55.76	67.75	77.00
SSRT [48]	74.00	99.75	0.00	**97.50**	21.18	60.00	58.74
SGLF (Ours)	**96.80**	**100.00**	**97.25**	50.50	85.65	60.50	**81.78**

IV. EXPERIMENTS

A. Datasets and Experimental Setup

1) Image Classification Datasets for DA: The NWPU-RESISC45 dataset [52] consists of 31 500 optical images from 45 categories. The MLRSNet dataset [53] contains 109 161 optical images of 46 categories. We select the NWPU-RESISC45 or MLRSNet dataset as the source domain. The WHU-SAR6 dataset [54] contains 2275 SAR images collected from GaoFen 3 [55] and Sentinel-1 [56], which have six categories: bridge, mountain, building, lake, farmland, and river. Here, we choose WHU-SAR6 as the target domain for the experiment.

To better study the DA of SAR images, we construct a new SAR dataset: NWPU-SAR6. The NWPU-SAR6 dataset contains 2644 SAR images of a high resolution of 1 m by collecting sample images from QXS-SAROPT [57], SAR-aircraft [58], and MSAR-1.0 [59], which comprises the following six categories: airplane, bridge, building, farmland, harbor, and storage-tank. Some examples of scene images are shown in Fig. 3. We chose NWPU-SAR6 as the target domain. In MLRSNet and NWPU-RESISC45 datasets, we choose the same categories as NWPU-SAR6 and WHU-SAR6, respectively, for DA classification from optical to SAR.

2) Object Detection Datasets for DA: The MASATI-ship dataset consists of 2368 satellite images related to the ship category as the source domain from the MASATI-v2 [61] dataset according to the data annotation. SSDD [62] is an SAR ship dataset, which consists of 1160 images as the target domain. MVRSD [63] is a dataset comprising 3000 vehicle images as the optical domain. SARDet-vehicle is an SAR dataset separated from the SARDet-100K [64] dataset, which consists of 940 images and 1791 vehicle instances as the target domain. DIOR-airplane is a dataset consisting of only 1387 airplane images selected from the DIOR [65] dataset as the source domain. SAR-airplane is an aircraft image dataset with a size of 640 × 640 obtained by the GaoFen-3 system, which includes 874 images as the target domain.

B. Implementation Details

We carry out the proposed method on the PyTorch [66] framework, and the GPU is NVIDIA 1080Ti. We take stochastic gradient descent (SGD) as the optimizer with a momentum of 0.9, a learning rate of 0.001, and a weight decay of 5×10^{-4}. For the backbone network, we utilize pretrained ResNet-50 [44] to initialize the backbone network. We employ the mean accuracy (Avg.) as an evaluation metric to evaluate the performance of the proposed method and some advanced methods. For weighting factors of $\mathcal{L}_{\text{PGAL}}$ and $\mathcal{L}_{\text{RPLL}}$, we fix $\lambda_1 = 0.1$ and $\lambda_2 = 0.01$ in our all experiments.

C. DA for Scene Classification

We compare our method with other advanced DA methods: ResNet-50 [44], DANN + E [45], CDAN + E [22], AFN [9], CDAN + BNM [46], ATM [10], CGDM [47], and SSRT [48]. Tables I and III show all the experimental results when the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.

TABLE II
ACCURACY (%) ON MLRSNET → NWPU-SAR6 FOR DA

Methods	airplane	building	farmland	harbor	storage-tank	bridge	Avg.
ResNet-50 [44]	33.00	51.40	44.44	60.58	23.28	26.60	39.88
DANN+E [45]	18.40	30.80	20.66	61.59	86.13	**97.78**	52.56
CDAN+E [22]	4.00	40.20	93.72	62.86	83.21	30.38	52.40
AFN [9]	3.60	33.80	81.82	80.95	60.58	27.27	48.00
CDAN+BNM [46]	14.20	94.71	70.79	56.93	50.55	43.60	55.13
ATM [10]	34.60	25.60	75.87	**83.81**	64.60	38.80	53.88
CGDM [47]	**57.60**	44.40	**97.19**	66.03	21.53	0.44	47.87
SSRT [48]	0.20	0.00	0.00	96.83	60.22	46.00	33.88
SGLF (Ours)	21.00	**95.21**	64.44	66.79	**93.35**	46.40	**64.53**

TABLE III
ACCURACY (%) ON MLRSNET → WHU-SAR6 FOR DA

Methods	bridge	mountain	building	lake	farmland	river	Avg.
ResNet-50 [44]	35.20	89.75	31.50	63.75	65.65	27.50	52.23
DANN+E [45]	**81.60**	96.25	80.75	48.50	48.00	60.50	69.27
CDAN+E [22]	63.20	88.00	93.50	70.75	33.88	50.00	66.56
AFN [9]	13.60	**99.50**	82.00	49.75	60.24	54.50	59.93
CDAN+BNM [46]	13.60	99.25	89.00	33.50	74.35	69.75	63.24
ATM [10]	48.80	96.75	83.50	26.50	68.94	73.50	66.33
CGDM [47]	24.80	99.25	**96.50**	59.00	49.18	**81.75**	68.41
SSRT [48]	55.20	99.50	90.75	**98.75**	4.00	72.25	70.08
SGLF (Ours)	37.20	92.75	92.25	69.75	**76.00**	73.75	**73.62**

WHU-SAR6 dataset is the target domain; Tables II and IV show the experimental results when the NWPU-SAR6 dataset is used as the target domain.

1) NWPU-RESISC45 → WHU-SAR6: We provide the results on the NWPU-RESISC45 → WHU-SAR6 adaptation datasets. As shown in Table I, compared with other DA methods, we can see that our proposed method achieves the optimal performance, outperforming the second-best CGDM [47] by an average of 4.78%. For each category, we can notice that the proposed method can alleviate the domain gap over some categories in WHU-SAR6, e.g., the bridge category gets 96.80%, the mountain category gets 100.00%, and the building category gets 97.25% as the best performance in Avg., which is highlighted in Table I. In contrast to CGDM, the proposed method achieved 9.2% and 0.25% improvements for the categories of bridge and mountain in WHU-SAR6, respectively. For the farmland and river categories, the category average accuracy is lower than CDAN + E [22], respectively. Nevertheless, our proposed method outperforms CDAN + E [22] in all other categories. This shows that SGLF has superior overall generalization ability, ensuring more stable and accurate recognition in different categories.

2) MLRSNet → NWPU-SAR6: From Table II, the proposed method achieved an average accuracy of 64.53%, which effectively improved the Source-only (no adaptation) model result by 24.65%. Beyond that, compared with existing methods, our method is better than DANN + E [45] of 11.97%, CDAN + E [22] of 12.13%, AFN [9] of 16.53%, CDAN + BNM [46] of 9.40%, ATM [10] of 10.65%, CGDM [47] of 16.65%, and SSRT [48] of 30.65% in terms of mean accuracy. For the building, farmland, harbor, and bridge categories, CDAN + BNM [46] achieves 94.71% in building, CGDM [47] achieves 97.19% in farmland, ATM [10] achieves 83.81% in harbor, and DANN + E [45] achieves 97.78% in bridge; the proposed method is comparable but slightly different in these four categories. Nevertheless, our method is still superior to other algorithms on the whole, which effectively verifies the effectiveness of the proposed method from the optical to the SAR domain.

3) MLRSNet → WHU-SAR6: In Table III, we record the experimental results of MLRSNet → WHU-SAR6. Compared to the Source-only classification method, we note that our method brings a significant increase of 21.39% in terms of Avg., which clearly proves the advantage of our method with better generalization ability under the DA setting. Besides, our method also better than DANN + E [45] by 4.35%, CDAN + E [22] by 7.06%, AFN [9] by 13.69%, CDAN + BNM [46] by 10.38%, ATM [10] by 7.29%, CGDM [47] by 5.21%, and SSRT [48] by 3.54% in terms of Avg., respectively. These significant improvements mean the rationality of the proposed method. The reasons for this remarkable growth are given as follows: 1) the URPG module is deployed to generate robust pseudolabels by modeling the uncertainty of correct labeling for each target-domain sample, which further alleviates the representation bias toward optical-annotated images and 2) the CSGA module can preserve and align the deep feature with second-order semantic dependencies by modeling the cross-modality visual semantic graph, which captures more informative commonalities between the optical and SAR images.

4) NWPU-RESISC45 → NWPU-SAR6: Table IV gives the results on NWPU-RESISC45 → NWPU-SAR6. Our final

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ACCURACY (%) ON NWPU-RESISC45 → NWPU-SAR6 FOR DA

Methods	airplane	building	farmland	harbor	storage-tank	bridge	Avg.
ResNet-50 [44]	33.20	74.71	70.79	21.90	53.44	30.80	47.47
DANN+E [45]	51.00	41.60	90.41	31.43	70.80	48.12	55.56
CDAN+E [22]	19.80	35.60	94.38	63.49	73.36	**60.00**	57.77
AFN [9]	40.40	26.80	**96.52**	27.30	62.41	45.90	49.84
ATM [10]	76.80	41.40	90.25	5.40	**88.69**	40.80	57.22
CGDM [47]	92.80	94.88	14.29	19.34	40.35	45.20	51.14
SSRT [48]	5.80	84.96	4.76	4.01	1.33	49.40	25.04
SGLF (Ours)	**93.80**	**98.68**	58.41	**64.60**	51.44	44.80	**68.62**

model yields 68.62% in terms of mean accuracy, which significantly improves the Source-only model result by 21.15%. Furthermore, our method is superior to the advanced method DANN + E [45] by 13.06%, CDAN + E [22] by 10.85%, AFN [9] by 18.78%, CDAN + BNM [46] by 0.009%, ATM [10] by 11.40%, CGDM [47] by 17.48%, and SSRT [48] by 43.58% in terms of mean accuracy. As shown in Table IV, although our method may not achieve optimal accuracy in certain categories, the average accuracy of our proposed method achieves the best experimental results overall and shows that our method is highly adaptable in scenarios ranging from optical to SAR domains.

In general, SGLF achieves competitive average accuracy on NWPU-RESISC45 → WHU-SAR6, MLRSNet → NWPU-SAR6, MLRSNet → WHU-SAR6, and NWPU-RESISC45 → NWPU-SAR6, demonstrating its robustness in feature alignment. For some categories, SGLF achieves the best classification in Tables I–IV; these achievements underline SGLF's capacity to capture domain-invariant semantic structures effectively across multiple categories. Some methods excel in isolated categories due to transformer-based representations or adversarial learning, such as the lake category in the SSRT [48] method (see Tables I and III), the bridge category in the DANN + E [45] (see Tables II and III), and the storage-tank in the ATM [10] method (see Table IV). Nevertheless, these comparison methods tend to lack consistent performance across all categories. In contrast, SGLF ensures a more balanced adaptation across a range of categories, providing stable and reliable recognition rather than excelling only in a few select cases. This is attributed to the fact that the URPG module generates robust pseudolabels by modeling the uncertainty of the correct labeling of each target-domain sample. With the assistance of these generated high-quality pseudolabels, the CSGA module achieves effective alignment of the semantic features of optical images and SAR images at the category level.

D. DA for Object Detection

1) MASATI-Ship → SSDD: Table V shows the detection results of MASATI-ship → SSDD. By plugging our method into D-adapt [60], the performance can be improved significantly. Here, we obtain a significant growth of 22.29% over the Source-only model. In feature-level adaptation methods, we hike up the mAP by 14.74% and 19.12% compared to SWDA [35] and ICCR-VDD [36], respectively, demonstrating the robustness of the proposed method. The main reasons for this performance improvement are given as follows.

1) Since CSGA utilizes the graph message-passing mechanism to perform cross-modality interaction on node information, it can mine high-level node information hidden in speckle noise in SAR images. In this way, the instance categories in the SAR image are aligned with the instance categories in the optical image, which greatly alleviates the misalignment problem of the instances.
2) Since the SAR image in the target domain has no labels and the SAR image contains speckle noise, this further enhances the difficulty of target detection in the SAR image. URPG dynamically optimizes the pseudolabels by modeling the uncertainty of correct labeling, which greatly improves the interference caused by speckle noise in SAR images so that the instance in the optical image can be correctly matched with the instance in the SAR image.

Fig. 4 shows detection examples from the SSDD dataset. We can see that compared with other methods, our method can alleviate the misclassification caused by speckle noise in SAR images and accurately locate and recognize objects. These further demonstrate that the combination of robust self-training and semantic graph alignment can indeed alleviate the impact of domain shift.

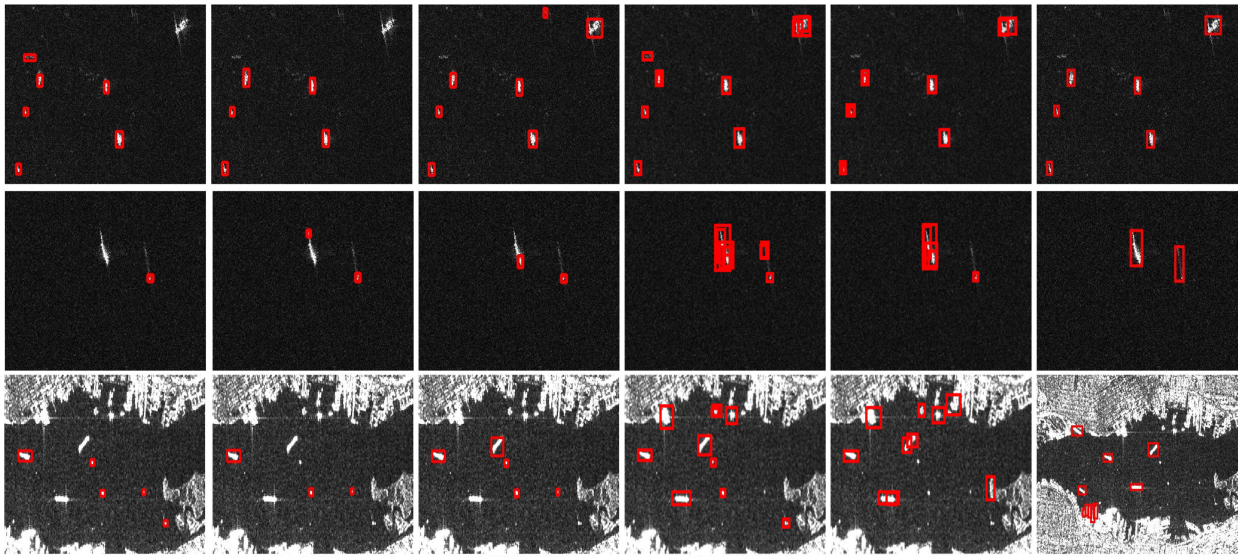
2) MVRSD → SARDet-Vehicle: In Table V, we also obtain 67.49% mAP on the DA task from MVRSD to SARDet-vehicle, which is the best result among all peers. The proposed method reaches a gain of 32.59% over the nonadaptive model in terms of mAP. Furthermore, we find that our method has some gains compared with ICCR-VDD [36] and D-adapt [60]. This increase in performance mainly comes from the collaboration of URPG and CSGA, which greatly alleviates the overadaptation problem between optical and SAR domains.

In addition, we provide detection examples from the SARDet-vehicle dataset. As shown in Fig. 5(a)–(c), the Source-only, SWDA [35], and ICCR-VDD [36] models only recognize some salient objects. From Fig. 5(d) and (e), it is noticeable that our method can impose more true positives to correct the detection error of objects. Even though our method exhibits missed detections, particularly in dense vehicle regions as evidenced by the ground-truth label in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.

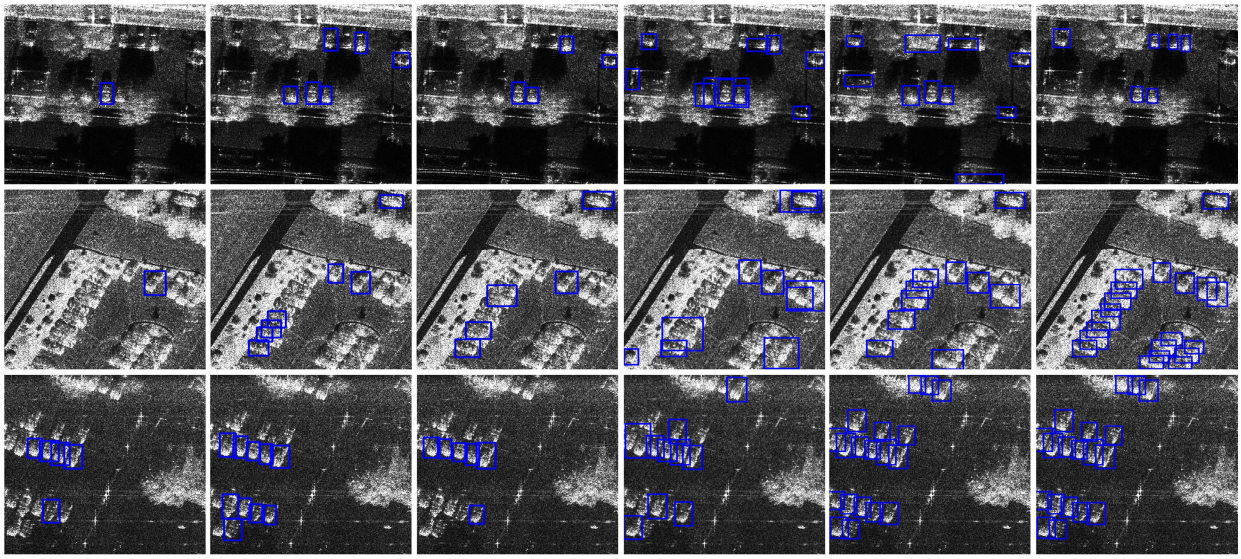
TABLE V
DETECTION RESULTS (%) ON MASATI-SHIP → SSDD, MVRSD → SARDET-VEHICLE, AND DIOR-AIRPLANE → SAR-AIRPLANE, RESPECTIVELY

Setting	MASATI-ship→SSDD	MVRSD→SARDet-vehicle	DIOR-airplane→SAR-airplane
Method	*ship* mAP	*vehicle* mAP	*airplane* mAP
Source-only	32.88	34.90	22.09
SWDA [35]	40.43	49.63	37.39
ICCR-VDD [36]	36.05	41.82	30.50
D-adapt [60]	49.79	60.97	39.31
SGLF (Ours)	**55.17**	**67.49**	**42.21**



(a) (b) (c) (d) (e) (f)

Fig. 4. Illustration of the detection results on MASATI-ship → SSDD. (a)–(f) Detection results of the Source-only model, the SWDA, the ICCR-VDD, the D-adapt, the proposed SGLF, and the ground-truth labels, respectively.



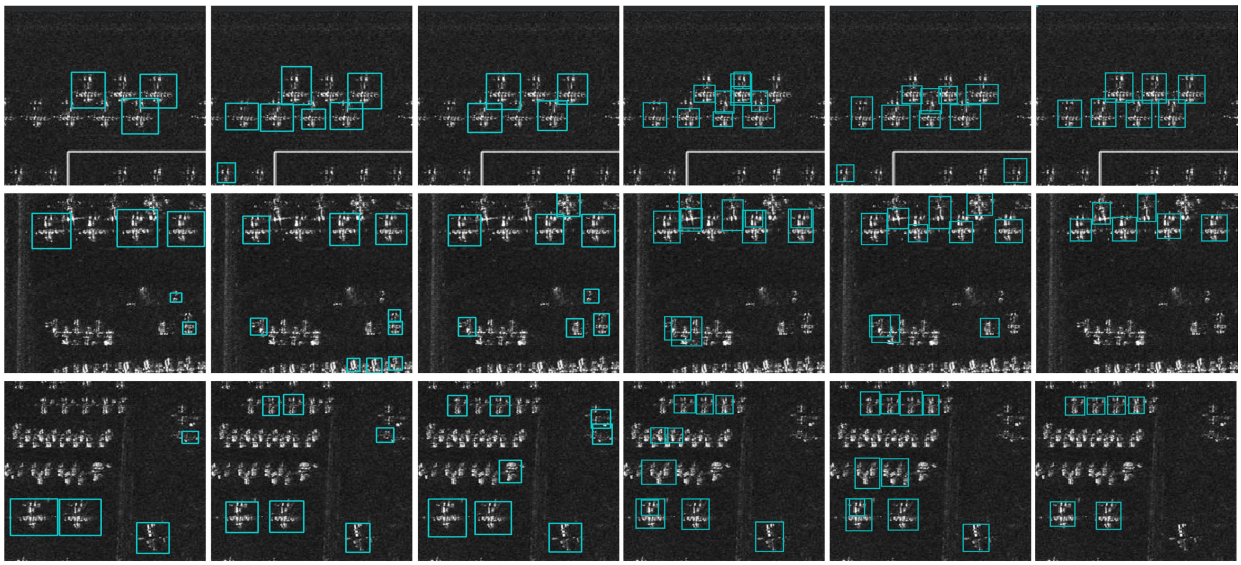
(a) (b) (c) (d) (e) (f)

Fig. 5. Illustration of the detection results on MVRSD → SARDet-vehicle. (a)–(f) Detection results of the Source-only model, the SWDA, the ICCR-VDD, the D-adapt, the proposed SGLF, and the ground-truth labels, respectively.

Fig. 5(f), it still demonstrates a significant capacity to mitigate domain shift, leading to improved performance on the target domain.

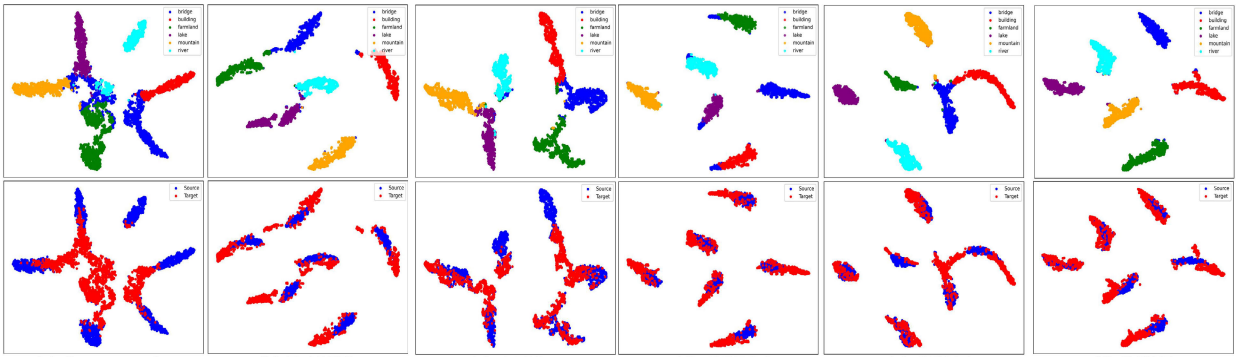
3) DIOR-Airplane → SAR-Airplane: Table V again displays the results of the DIOR-airplane to SAR-airplane task. We observe that our method obtains 42.21% mAP detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e) (f)

Fig. 6. Illustration of the detection results on DIOR-airplane → SAR-airplane. (a)–(f) Detection results of the Source-only model, the SWDA, the ICCR-VDD, the D-adapt, the proposed SGLF, and the ground-truth labels, respectively.



(a) (b) (c) (d) (e) (f)

Fig. 7. Visualization of features using t-SNE. The first row shows category information visualization, where different colors represent different categories. The second row presents domain information visualization, where blue indicates source-domain samples and red indicates target-domain samples. Feature distribution results of the model trained by (a) source images only, (b) DANN + E, (c) AFN, (d) CGDM, (e) SSRT, and (f) SGLF, respectively. We take MLRSNet → WHU-SAR6 as an example.

accuracy that surpasses the existing comparison methods. In particular, the results of our method are improved by 2.90% on average compared to the baseline D-adapt [60], which clearly demonstrates the importance of our method. In contrast to SWDA [35] and ICCR-VDD [36], our method explores the discrepancy in interdomain and intradomain to enable knowledge transfer from optical to SAR, showing the effectiveness of our semantic graph alignment with the help of high-quality pseudolabels.

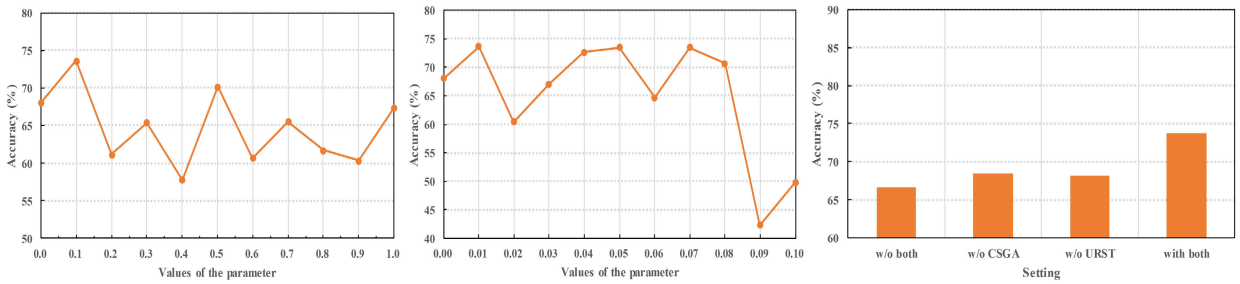
Furthermore, Fig. 6 shows detection adaptation examples from the SAR-airplane dataset. As shown in the third row of Fig. 6, we can see that our method can precisely detect the objects in the image. In Fig. 6(a)–(c), most of the airplane objects are not detected. Besides, our approach also eliminates the error of target information loss compared with the baseline D-adapt [60], showing the advantage in class information-guided semantic graph adaptation with the help of the SGLF framework. As illustrated in Fig. 6(e) and (f), the proposed uncertainty-based pseudolabel generator can produce a larger quantity of high-quality pseudolabels. However, a significant gap still exists between the quality of the generated pseudolabels and the ground-truth labels, especially in scenarios with dense airplanes or small objects. Even so, the framework develops a CSGA module to model and align deep second-order semantic dependence features through the generated high-quality pseudolabels, which effectively alleviates the domain bias of the detector.

E. Model Analysis

In this section, we further analyze the characteristics of the model in feature visualization, hyperparameter sensitivity, ablation study, and discussion.

1) Feature Visualization: To better understand the proposed method and prove the effectiveness of our model, we visualize the distribution of features learned from the model by t-SNE [67]. We take MLRSNet → WHU-SAR6 as an example.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 8. (a) Sensitivity analysis of parameter λ_1. (b) Sensitivity analysis of parameter λ_2. (c) Ablation study on MLRSNet → WHU-SAR6; w/o is short for without.

TABLE VI

COMPARISON OF THE TRAINING EFFICIENCY AND INFERENCE EFFICIENCY WITH RESNET-101 BACKBONE UNDER MVRSD → SARDET-VEHICLE ADAPTATION SETTING

Methods	Training time (s/iter)	Inference time (s/image)
SWDA [35]	**0.3557**	0.0996
ICCR-VDD [36]	0.5740	0.0996
D-adapt [60]	1.5250	**0.0665**
SGLF (Ours)	1.050	**0.0665**

As shown in Fig. 7, the first row shows the visualization of category information, and the second row shows the visualization of domain information. It indicates that in any scenario, the optical domain samples exhibit a discriminative distribution. However, without DA, the SAR domain samples take the form of a cluttered distribution, with features from different categories overlapping, as shown in Fig. 7(a). As shown in Fig. 7(b)–(e), although some methods alleviate the misalignment problem for certain categories, such as the mountain category in Fig. 7(c) and the building category in Fig. 7(d), the misalignment problem between other categories is still serious overall. In Fig. 7(f), compared to the visualizations in Fig. 7(b)–(e), the t-SNE visualization results obtained by the SGLF method show that the farmland category is clearly separated, and the overlapping areas of other categories are relatively small. Therefore, SGLF ensures a more balanced adaptation across a range of categories, providing stable and reliable recognition rather than excelling only in a few select cases, validating the generalization ability and effectiveness of the proposed method.

2) Sensitivity to Hyperparameter: We provide the sensitivity analysis for balancing parameters of cross-domain category alignment loss and in-domain category alignment loss. Our model mainly involves two hyperparameters λ_1 and λ_2. We first fix λ_1 as 0.1 because this value is empirically found to make our method perform well. Then, we show the accuracy of MLRSNet → WHU-SAR6 by turning λ_2 from 0 to 0.1, which is shown in Fig. 8(b). After fix λ_2, we turning λ_1 from 0 to 1.0, which is shown in Fig. 8(a). We achieve the best accuracy is 73.62% when $\lambda_1 = 0.1$ and $\lambda_2 = 0.01$.

3) Ablation Study: To study the importance of each module in the SGLF framework, we carry on an ablation study on DA from optical to SAR. Our framework is composed of CSGA and URPG. To verify the impact of each module, we conduct experiments to remove CSGA and URPG, respectively. Take the average accuracy of MLRSNet → WHU-SAR6 as an example. As can be seen from Fig. 8(c), in contrast to adversarial learning, both modules significantly improve the performance of the model, and when they work together in coordination, the accuracy of the model is better. It can be clearly seen that our method can effectively alleviate the mismatch between optical and SAR image category instances compared with the vanilla adversarial learning method.

F. Discussion

In Tables I and III, when the target domain is WHU-SAR6, an average accuracy of 81.75% is obtained on NWPU-RESISC45 → WHU-SAR6, and an average accuracy of 73.62% is obtained on MLRSNET → WHU-SAR6. In Tables II and IV, when the target domain is NWPU-SAR6, the average accuracy obtained on MLRSNET → NWPU-SAR6 is 64.53%, and the average accuracy obtained on NWPU-RESISC45 → NWPU-SAR6 is 68.62%. The above experiments on DA scene classification show that when the target-domain SAR dataset is fixed and the source domain is an optical remote sensing dataset from different scenes, the performance of the model varies to some extent. When the source-domain optical remote sensing dataset is fixed and the target domain is the SAR dataset in different scenarios, the performance of the model also varies to some extent. Table V shows the DA object detection experiments from optical to SAR in different scenarios, which indicates that the model's performance may vary under different objects and scenes. Therefore, the DA scene classification in Tables I–IV and the DA object detection experiment in Table V indicate that the proposed model can achieve DA from optical images to SAR images to a certain extent. Based on the above analysis, although domain transfer models corresponding to different targets and scenes are all different, our method can be used for target classification and detection from optical RSIs to SAR images. Therefore, our method is universal and can adapt to different scenes and targets.

G. Model Efficiency

As illustrated in Table VI, we conduct a comparative analysis of training time and inference speed (the inference time per image) between our proposed method and other state-of-the-art approaches. We employ a ResNet-101 backbone to perform

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.

experiments from MVRSD to the SARDet-vehicle dataset. As evidenced in the table, our proposed method achieves the most rapid inference speed of 0.0665 s/image. While our proposed method does not attain the optimal training time, its average training time does not exhibit a significant increase compared to other DA models. Consequently, the proposed framework offers a compelling efficiency.

V. Conclusion

In this article, we propose a novel SGLF to address the issue of DA from optical to SAR modalities. Our framework involved two innovative modules: the cross-modality graph alignment (CSGA) module and the URPG module. CSGA models and aligns the deep feature with second-order semantic dependencies by constructing a cross-modality semantic visual graph for better model generalization. URPG alleviates the bias in favor of representations derived from optical-annotated images to generate robust pseudolabels by modeling the uncertainty of correct labeling for each SAR image. These two critical modules are very well-aligned with each other, and their integration significantly amplifies the capabilities of semantic graph learning from two separate angles. Extensive experimentation demonstrates that our proposed method outperforms state-of-the-art approaches on scene classification (NWPU-RESISC45 $\rightarrow$ WHU-SAR6, MLRSNet $\rightarrow$ NWPU-SAR6, MLRSNet $\rightarrow$ NWPU-SAR6, and NWPU-RESISC45 $\rightarrow$ NWPU-SAR6) and object detection (MASATI-ship $\rightarrow$ SSDD, MVSRD $\rightarrow$ SARDet-vehicle, and DIOR-airplane $\rightarrow$ SAR-airplane) tasks. Moreover, we build an NWPU-SAR6 dataset, a valuable resource for researchers working on DA for SAR image scene classification.

References

[1] M. Datcu, Z. Huang, A. Anghel, J. Zhao, and R. Cacoveanu, "Explainable, physics-aware, trustworthy artificial intelligence: A paradigm shift for synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 1, pp. 8–25, Mar. 2023.

[2] Y. Shi, L. Du, and Y. Guo, "Unsupervised domain adaptation for SAR target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6372–6385, 2021.

[3] Z. Huang, C. O. Dumitru, Z. Pan, B. Lei, and M. Datcu, "Classification of large-scale high-resolution SAR images with deep transfer learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 107–111, Jan. 2021.

[4] Z. Han, Z. Zhang, S. Zhang, G. Zhang, and S. Mei, "Aerial visible-to-infrared image translation: Dataset, evaluation, and baseline," *J. Remote Sens.*, vol. 3, p. 0096, Jan. 2023.

[5] L. Shi, R. Zhao, B. Pan, Z. Zou, and Z. Shi, "Unsupervised multimodal remote sensing image registration via domain adaptation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5626211.

[6] Z. Han, S. Zhang, Y. Su, X. Chen, and S. Mei, "DR-AVIT: Toward diverse and realistic aerial visible-to-infrared image translation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5004213.

[7] Z. Huang, Y. Liu, X. Yao, J. Ren, and J. Han, "Uncertainty exploration: Toward explainable SAR target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2023, Art. no. 4202314.

[8] Y. Liu, Y. Zhang, Y. Wang, and S. Mei, "Rethinking transformers for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617515.

[9] R. Xu, G. Li, J. Yang, and L. Lin, "Larger norm more transferable: An adaptive feature norm approach for unsupervised domain adaptation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1426–1435.

[10] J. Li, E. Chen, Z. Ding, L. Zhu, K. Lu, and H. T. Shen, "Maximum density divergence for domain adaptation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 11, pp. 3918–3930, Nov. 2021.

[11] Y. Shi, L. Du, Y. Guo, and Y. Du, "Unsupervised domain adaptation based on progressive transfer for ship detection: From optical to SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5230317.

[12] J. Zhang, S. Li, Y. Dong, B. Pan, and Z. Shi, "Hierarchical similarity alignment for domain adaptive ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5240611.

[13] Y. Yuan, Z. Rao, C. Lin, Y. Huang, and X. Ding, "Adaptive ship detection from optical to SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[14] Y. Shi, L. Du, C. Li, Y. Guo, and Y. Du, "Unsupervised domain adaptation for SAR target classification based on domain- and class-level alignment: From simulated to real data," *ISPRS J. Photogramm. Remote Sens.*, vol. 207, pp. 1–13, Jan. 2024.

[15] B. Zou, J. Qin, and L. Zhang, "Cross-scene target detection based on feature adaptation and uncertainty-aware pseudo-label learning for high resolution SAR images," *ISPRS J. Photogramm. Remote Sens.*, vol. 200, pp. 173–190, Jun. 2023.

[16] S. Zhao, Y. Zhang, Y. Luo, Y. Kang, and H. Wang, "Dynamically self-training open set domain adaptation classification method for heterogeneous SAR image," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[17] D. Xiang, X. Pan, H. Ding, J. Cheng, and X. Sun, "Two-stage registration of SAR images with large distortion based on superpixel segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5211115.

[18] D. Xiang, H. Ding, X. Sun, J. Cheng, C. Hu, and Y. Su, "PolSAR image registration combining Siamese multiscale attention network and joint filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5208414.

[19] X. Sun, X. Li, D. Xiang, and C. Hu, "SAR vehicle image generation with integrated deep imaging geometric information," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 132, Aug. 2024, Art. no. 104028.

[20] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and J. W. Vaughan, "A theory of learning from different domains," *Mach. Learn.*, vol. 79, no. 1, pp. 151–175, 2010.

[21] M. Long, H. Zhu, J. Wang, and M. I. Jordan, "Deep transfer learning with joint adaptation networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 2208–2217.

[22] M. Long, Z. Cao, J. Wang, and M. I. Jordan, "Conditional adversarial domain adaptation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, Jan. 2017, pp. 1645–1655.

[23] W. Zellinger, T. Grubinger, E. Lughofer, T. Natschläger, and S. Saminger-Platz, "Central moment discrepancy (CMD) for domain-invariant representation learning," 2017, *arXiv:1702.08811*.

[24] I. J. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[25] X. Zhang, X. Yao, X. Feng, G. Cheng, and J. Han, "DFENet for domain adaptation-based remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5611611.

[26] J. Zhang, J. Liu, B. Pan, and Z. Shi, "Domain adaptation based on correlation subspace dynamic distribution alignment for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7920–7930, Nov. 2020.

[27] Z. Huang, Z. Pan, and B. Lei, "What, where, and how to transfer in SAR target recognition based on deep CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2324–2336, Apr. 2020.

[28] X. Qian, C. Li, W. Wang, X. Yao, and G. Cheng, "Semantic segmentation guided pseudo label mining and instance re-detection for weakly supervised object detection in remote sensing images," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 119, May 2023, Art. no. 103301.

[29] N. Ammour, L. Bashmal, Y. Bazi, M. M. A. Rahhal, and M. Zuair, "Asymmetric adaptation of deep features for cross-domain classification in remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 597–601, Apr. 2018.

[30] X. Deng, Y. Zhu, Y. Tian, and S. Newsam, "Scale aware adaptation for land-cover classification in remote sensing imagery," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 2160–2169.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.

[31] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.

[32] G. Yang and H. Lang, "Semisupervised heterogeneous domain adaptation via dynamic joint correlation alignment network for ship classification in SAR imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[33] Q. He, L. Zhao, K. Ji, and G. Kuang, "SAR target recognition based on task-driven domain adaptation using simulated data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[34] Y. Xu, H. Lang, L. Niu, and C. Ge, "Discriminative adaptation regularization framework-based transfer learning for ship classification in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 11, pp. 1786–1790, Nov. 2019.

[35] K. Saito, Y. Ushiku, T. Harada, and K. Saenko, "Strong-weak distribution alignment for adaptive object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6956–6965.

[36] A. Wu, R. Liu, Y. Han, L. Zhu, and Y. Yang, "Vector-decomposed disentanglement for domain-invariant object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9342–9351.

[37] L. Zhao and L. Wang, "Task-specific inconsistency alignment for domain adaptive object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14197–14206.

[38] S. Li, J. Huang, X.-S. Hua, and L. Zhang, "Category dictionary guided unsupervised domain adaptation for object detection," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, no. 3, pp. 1949–1957.

[39] Y. Zhu, X. Sun, W. Diao, H. Wei, and K. Fu, "DualDA-net: Dual-head rectification for cross-domain object detection of remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5612616.

[40] S. Zhao, Y. Luo, T. Zhang, W. Guo, and Z. Zhang, "A domain specific knowledge extraction transformer method for multisource satellite-borne SAR images ship detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 16–29, Apr. 2023.

[41] Z. Zhang, P. Cui, and W. Zhu, "Deep learning on graphs: A survey," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 1, pp. 249–270, Mar. 2020.

[42] W. Li, X. Liu, and Y. Yuan, "SIGMA: Semantic-complete graph matching for domain adaptive object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5281–5290.

[43] R. Sinkhorn, "A relationship between arbitrary positive matrices and doubly stochastic matrices," *Ann. Math. Statist.*, vol. 35, no. 2, pp. 876–879, Jun. 1964.

[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[45] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in *Proc. 32nd Int. Conf. Mach. Learn. (ICML)*, vol. 2, Jul. 2015, pp. 1180–1189.

[46] S. Cui, S. Wang, J. Zhuo, L. Li, Q. Huang, and Q. Tian, "Towards discriminability and diversity: Batch nuclear-norm maximization under label insufficient situations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 3941–3950.

[47] Z. Du, J. Li, H. Su, L. Zhu, and K. Lu, "Cross-domain gradient discrepancy minimization for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3937–3946.

[48] T. Sun, C. Lu, T. Zhang, and H. Ling, "Safe self-refinement for transformer-based domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 7191–7200.

[49] E. Arazo, D. Ortego, P. Albert, N. O'Connor, and K. McGuinness, "Unsupervised label noise modeling and loss correction," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 312–321.

[50] Z. Ma and A. Leijon, "Bayesian estimation of beta mixture models with variational inference," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 11, pp. 2160–2173, Nov. 2011.

[51] A. Ghosh, H. Kumar, and P. S. Sastry, "Robust loss functions under label noise for deep neural networks," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2017, vol. 31, no. 1, pp. 1919–1925.

[52] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[53] X. Qi et al., "MLRSNet: A multi-label high spatial resolution remote sensing dataset for semantic scene understanding," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 337–350, Nov. 2020.

[54] B. Su, J. Liu, X. Su, B. Luo, and Q. Wang, "CFCANet: A complete frequency channel attention network for SAR image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11750–11763, 2021.

[55] Q. Zhang, "System design and key technologies of the GF-3 satellite," *Acta Geodaetica et Cartographica Sinica*, vol. 46, no. 3, p. 269, Mar. 2017.

[56] R. Torres et al., "GMES Sentinel-1 mission," *Remote Sens. Environ.*, vol. 120, pp. 9–24, May 2012.

[57] M. Huang et al., "The QXS-SAROPT dataset for deep learning in SAR-optical data fusion," 2021, *arXiv:2103.08259*.

[58] P. Zhang et al., "SEFEPNet: Scale expansion and feature enhancement pyramid network for SAR aircraft detection with small sample dataset," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3365–3375, 2022.

[59] R. Xia et al., "CRTransSar: A visual transformer based on contextual joint representation learning for SAR ship detection," *Remote Sens.*, vol. 14, no. 6, p. 1488, Mar. 2022.

[60] J. Jiang, B. Chen, J. Wang, and M. Long, "Decoupled adaptation for cross-domain object detection," 2021, *arXiv:2110.02578*.

[61] A.-J. Gallego, A. Pertusa, and P. Gil, "Automatic ship classification from optical aerial images with convolutional neural networks," *Remote Sens.*, vol. 10, no. 4, p. 511, Mar. 2018.

[62] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.

[63] D. Bai, Y. Yu, L. Song, B. Cheng, and H. Gao, "Construction and validation of remote sensing image dataset for fine-grained detection of military vehicles," *J. Image Graph.*, vol. 29, no. 12, pp. 3564–3577, 2024.

[64] Y. Li et al., "SARDet-100K: Towards open-source benchmark and toolkit for large-scale SAR object detection," 2024, *arXiv:2403.06534*.

[65] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[66] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, Jan. 2019, pp. 1–11.

[67] L. Van der Maaten and G. E. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, Jan. 2008.



Xiufei Zhang received the M.S. degree from Northwestern Polytechnical University, Xi'an, China, in 2018, where he is currently pursuing the Ph.D. degree.

His research interests include computer vision and remote sensing image processing, especially object detection, scene classification, and semantic segmentation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.



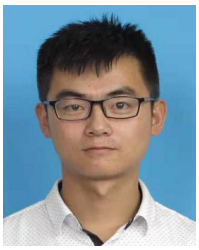
Zhongling Huang (Member, IEEE) received the B.Sc. degree from Beijing Normal University, Beijing, China, in 2015, and the joint Ph.D. degree from the University of Chinese Academy of Sciences (UCAS), Beijing, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, in 2020.

She was a Visiting Scholar at German Aerospace Center, Wessling, Germany, from 2018 to 2019. She is currently an Associate Professor with the BRain and Artificial INtelligence Laboratory (BRAIN LAB), School of Automation, Northwestern Polytechnical University, Xi'an, China. Her research interests include explainable deep learning for synthetic aperture radar (XAI4SAR), SAR image interpretation, deep learning, and remote sensing data mining.



Xiwen Yao (Member, IEEE) received the B.S. and Ph.D. degrees from Northwestern Polytechnical University, Beijing, China, in 2010 and 2016, respectively.

He is currently a Professor with Northwestern Polytechnical University. His research interests include computer vision and remote sensing image processing, especially fine-grained image classification and object detection.



Xiaoxu Feng received the Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 2023.

He is an Associate Professor with the Institute of Artificial Intelligence, Hefei Comprehensive National Science Center, Hefei, China. His research interests include computer vision and deep learning, especially object detection and scene classification.



Gong Cheng (Senior Member, IEEE) received the B.S. degree from Xidian University, Xi'an, China, in 2007, and the M.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, in 2010 and 2013, respectively.

He is currently a Professor with Northwestern Polytechnical University. His main research interests include computer vision and pattern recognition.



Junwei Han (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees in pattern recognition and intelligent systems from Northwestern Polytechnical University, Xi'an, China, in 1999, 2001, and 2003, respectively.

He is currently a Professor with Northwestern Polytechnical University. His research interests include computer vision and brain-imaging analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:59:03 UTC from IEEE Xplore. Restrictions apply.